THE UNIVERSITY of EDINBURGH
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Watching Coffee Dry

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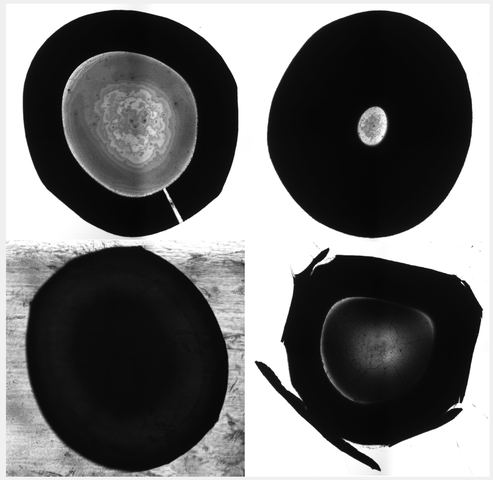


Figure 1. **Coffee ring's formed with a dried silica solution (particle diameter $\phi = 0.3\mu\mathrm{m}$, mass fraction $\mathbf{m_f = 0.0400(3)}$) on a range of substrates.** Images produced by an Olympus BX50 optical microscope. The diameters of the rings are $\sim 1mm$ on the solid-liquid interface.

Introduction

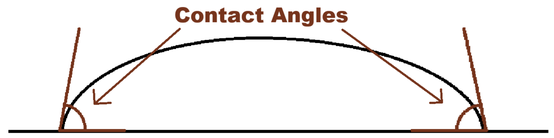
If you have ever left spilt coffee out to dry you will have noticed that it leaves behind a concentrated ring of coffee particles in its wake. The effect occurs in all colloidal fluids and plays an important role in the inkjet printing industry where consistent drying of inks is required to produce legible characters across a range of substrates [1]. In recent years, this effect has garnered lots of attention in the literature after it's initial disovery in 1997 [2]. Few studies have attempted to correlate the coffee ring deposition with the interfacial tensions of a droplet in a three-phase system; this project aims to fill that research gap.

The Contact Angle

The initial contact angle is a convenient analogue to gather information about the interfacial tensions in a three-phase vapour-solid-gas system. The quantities are linked by Young's equation [3]:

$$\cos\theta = \frac{\gamma_{SV} - \gamma_{SL}}{\gamma_{LV}}$$

where θ is the contact angle and γ are the interfacial tensions for the respective phase boundaries. An illustration of a contact angle is presented below:



Measurement of the initial contact angle requires only a camera of sufficient resolution and a software package for analysing the images. This approach is orders of magnitude less complicated than attempting to measure all three of the interfacial tensions directly.

Classifying the Coffee Ring

To quantitatively describe the coffee ring particle deposition the ratio of it's outer diameter to width was used:

$$R(\theta) = \frac{\phi_{outer}}{\phi_{outer} - \phi_{inner}}$$

where θ is the initial contact angle for the substrate and ϕ represents diameter measurements of the coffee ring. This quantity is independent of length scale which is convenient because pixel values do not need to be converted to real length scales.

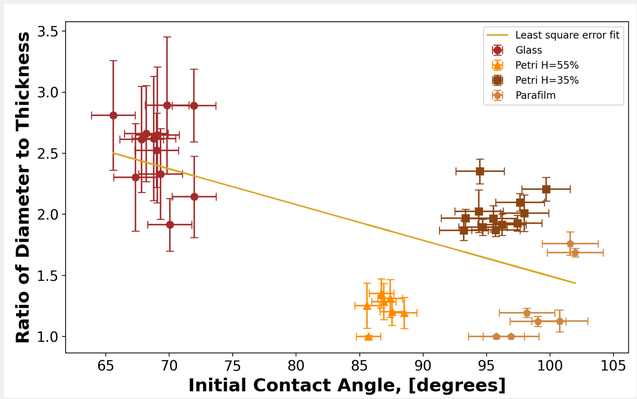


Figure 2. **Plot of coffee ring outer diameter to width ratio against initial contact angle for various substrates.** Petri film samples were conducted at two different relative humidities, indicated in the legend.

Trends in the Data

A weak negative correlation between initial contact angle, θ, and ratio $R(\theta)$ is presented in Figure 2 for a range of substrates. This trend is expected from measurements made in [4]. The humidity was uncovered as an important factor which was overlooked in this experiment. Further research into this topic should thus be performed at a constant relative humidity, as was illuminated in [5].

References

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2. Deegan, R. D. *et al.* Capillary flow as the cause of ring stains from dried liquid drops. *Nature.* **389.** 827–829. 1997.
3. Bico, J., Marzolin, C. & Quéré, D. Pearl drops. *EPL : a letters journal exploring the frontiers of physics.* **47.** 220–226. 1999.
4. Yi, J., Jeong, H. & Park, J. Modulation of nanoparticle separation by initial contact angle in coffee ring effect. *Micro and Nano Systems Letters.* **6.** 17. 2018.
5. Chhasatia, V. H., Joshi, A. S. & Sun, Y. Effect of relative humidity on contact angle and particle deposition morphology of an evaporating colloidal drop. *Applied Physics Letters.* **97.** 231909. 2010.